# Braking System Design Report

## I. Rulebook Constraints

According to the IKR 2025 rulebook:

- Karts must have a hydraulic braking system.
- On applying brakes, all four wheels should lock up.
- The kart must safely stop after accelerating at least up to 25 km/h during the brake test.
- The tire specification to be followed is in the following table:

| | Type | Category | Dimension |
|---|---|---|---|
| Dry (slick) tires | Front | D1 | 4.5×10.0-5 |
| Wet tires | Front | W1 or W2 | 4.5×10.0-5 |
| Dry (slick) tires | Rear | D1 | 7.1×11.0-5 |
| Wet tires | Rear | W1 or W2 | 6.0×11.0-5 |

TABLE I: Tire specifications

## II. Force Consideration

- Maximum foot force applied by driver: 250 N
- Pedal mechanical advantage: 5.5

Effective force transmitted to master cylinder:

$$F_{MC} = F_{foot} \times MA = 250 \times 5.5 = 1375\,N$$

## III. Working Principle

The brake pedal, using a mechanical linkage, actuates the clutch master cylinder:

The rear braking circuit operates one caliper on the rear axle.

When the pedal is pressed, hydraulic pressure develops inside master cylinder actuates the caliper piston, pressing the brake pads against the rotating discs to generate braking torque.

## IV. Design Considerations

Key aspects considered during design:

- Sufficient hydraulic pressure generation within safe limits.
- Adequate braking torque to decelerate from 40 km/h.
- Avoidance of component failure under maximum force.

## V. Component Selection

TABLE II: Selected Components and Specifications

| **Component** | **Specification** |
|---|---|
| Master Cylinder | 21.9 mm Bore (x1 units) |
| Rear Caliper | 32 mm Bore (x1 unit) |
| Rear Brake Disc | 130 mm ID, 200 mm OD |
| Rear Tyre Diameter | 279.4 mm |
| Rear Brake Pad Area | 1800 $mm^2$ |
| Pedal Mechanical Advantage | 5.5 |

## VI. Pictures of Design and Components

Fig. 1: Images of Brake pad

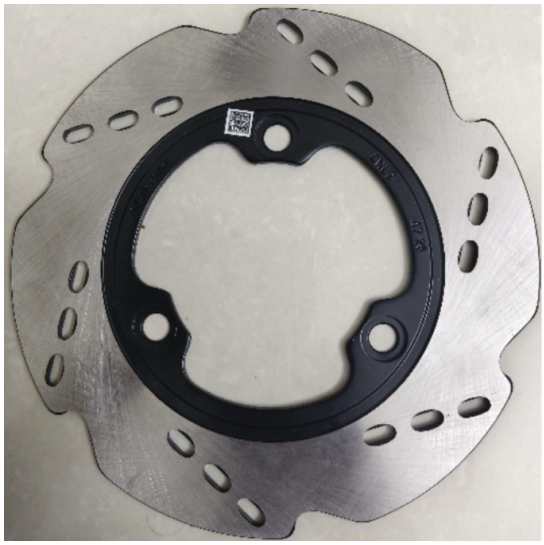

Fig. 2: Images of Brake disc

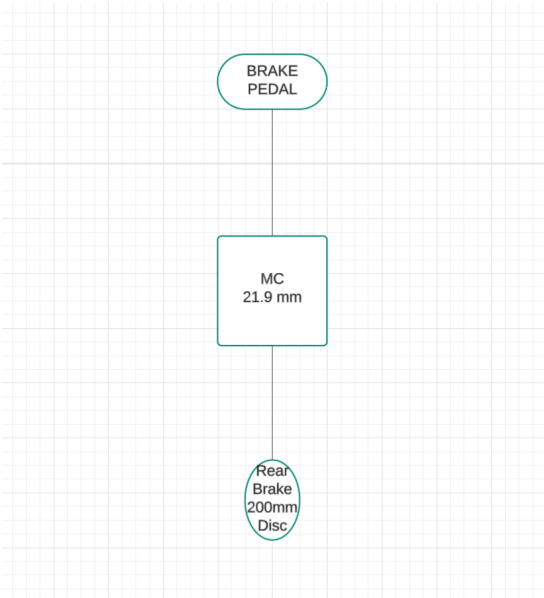


Fig. 3: FLOW CHART

## VII. Part Analysis

Currently, no finite element analysis (FEA) is conducted on brake discs or pedal arms. Static load analysis and thermal analysis are planned in future work.

## VIII. Formulation and Calculations

### A. Master Cylinder Area

Bore diameter of master cylinder, $D_{MC} = 21.9\,mm = 0.0219\,m$

Radius:

$$r_{MC} = \frac{D_{MC}}{2} = 0.01095\,m$$

Area:

$$A_{MC} = \pi r_{MC}^2 = \pi(0.01095)^2 = 3.769 \times 10^{-4}\,m^2$$

### B. Hydraulic Pressure

Effective force on master cylinder:

$$F_{MC} = 1375\,N$$

Hydraulic Pressure generated:

$$\begin{aligned} P &= \frac{F_{MC}}{A_{MC}} \\ &= \frac{1375}{3.769 \times 10^{-4}} \\ &= 3.65 \times 10^6\,\text{Pa} = 3.65\,\text{MPa} \end{aligned}$$

### C. Caliper Piston Forces

**Rear Caliper** (Bore 32 mm):

Radius:

$$r_{Rear-Caliper} = 16\,mm = 0.016\,m$$

Area:

$$A_{Rear-Caliper} = \pi(0.016)^2 = 8.042 \times 10^{-4}\,m^2$$

Force:

$$F_{Rear-Caliper} = 3.65 \times 10^6 \times 8.042 \times 10^{-4} = 2935.3\,N$$

### D. Braking Torque Generation

Disc effective radii:

Rear Disc:

$$r_{disc-rear} = \frac{130 + 200}{4} = 82.5\,mm = 0.0825\,m$$

**Frictional Force** between pad and disc: Coefficient of friction $\mu_{pad-disc} = 0.35$

Rear:

$$F_{Friction-Rear} = 0.35 \times 2935.3 = 1027.36\,N$$

**Braking Torque**:

Rear Torque (each wheel):

$$T_{Rear} = F_{Friction-Rear} \times r_{disc-rear} = 1027.36 \times 0.0825 = 84.75\,Nm$$

### E. Total Braking Force

Front wheel radius = 0.127 m Rear wheel radius = 0.1397 m

Force at tyre contact patch:

Rear:

$$F_{Rear-Tyre} = \frac{84.75}{0.1397} = 606.7\,N$$

Total Braking Force:

$$F_{Total-Braking} = F_{Rear-Tyre} = 606.7\,N$$

### F. Stopping Distance and Time Calculation

Initial Speed:

$$v = 40\,km/h = 11.11\,m/s$$

Mass of kart:

$$m = 180\,kg$$

Deceleration:

$$a = \frac{F_{Total-Braking}}{m} = \frac{606.7}{180} = 3.3705\,m/s^2$$

Braking time:

$$t_{braking} = \frac{v}{a} = \frac{11.11}{3.3705} = 3.296\,s$$

Braking distance:

$$d_{braking} = \frac{v^2}{2a} = \frac{(11.11)^2}{2 \times 3.3705} = 18.31\,m$$

### G. Stopping Distance During Reaction Time

Here, we calculate the distance traveled during the driver's reaction time, considering deceleration due to tyre-road friction.

$$d = v \times t + \frac{1}{2}at^2$$

Where:

- $v = 11.11\,m/s$ is the initial velocity.
- $t = 0.5\,s$ is the reaction time.
- $a = -\mu \times g = -0.7 \times 9.81 = -6.867\,m/s^2$ is the deceleration, where $\mu = 0.7$ is the coefficient of friction between tyre and road.

Substituting the values:

$$d = 11.11 \times 0.5 + \frac{1}{2} \times (-6.867) \times (0.5)^2$$

Simplifying:

$$d = 5.555 - 0.858 = 4.697 \,\text{meters}$$

Thus, the distance traveled during the reaction time is approximately 4.70 meters. Total stopping distance:

$$d_{total} = d_{braking} + d_{reaction} = 18.31 + 4.697 = 23.007\, m$$

## IX. Tabulations

TABLE III: Braking Parameters Summary

| Parameter | Value |
|---|---|
| Stopping Distance | 23.007 meters |
| Stopping Time | 3.796 seconds |
| Driver Foot Force | 250 N |
| Force at Master Cylinder | 1375 N |
| Hydraulic Pressure | 3.65 MPa |
| Total Braking Force | 606.7 N |

## X. Results

- Total Stopping Distance: **23.007 meters**
- Total Stopping Time: **3.796 seconds** (including reaction time)
- Hydraulic pressure generated: **3.65 MPa**
- Driver effort required: **250 N**
- The above calculations were also done for wet conditions, and considering wet tires, the results are tabulated in Table IV.

TABLE IV: Results for Wet Conditions

| **Parameter** | **Value** |
|---|---|
| Initial Velocity | 11.11 m/s (40 km/h) |
| Coefficient of Friction (Wet) | 0.3 |
| Deceleration | 2.943 m/s$^2$ |
| Reaction Time | 0.5 seconds |
| Reaction Distance | 4.697 meters |
| Braking Time | 3.77 seconds |
| Braking Distance | 20.97 meters |
| Total Stopping Time | 4.27 seconds |
| Total Stopping Distance | 25.667 meters |

## XI. Results

The braking system meets the safety requirements even under a worst-case scenario of braking from 40 km/h. Hydraulic pressures and forces are within component limits. Stopping distance and stopping time are within safe ranges, ensuring reliability of the system.